

RUZAEVKA, Kazakhstan

WMO: 289660

Lat: 52.8224N

Lon: 66.9581E

Elev: 745

StdP: 14.30

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.5	-21.7	-33.5	0.8	-27.1	-27.8	1.2	-21.5	32.3	17.7	31.7	17.7	2.1	230	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	89.5	65.6	86.2	64.9	82.8	63.8	68.5	83.0	67.0	81.2	65.5	79.0	10.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.0	91.6	73.0	62.1	85.7	71.9	60.3	80.3	70.9	33.2	83.2	31.9	81.0	30.7	79.3	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
31.7	25.6	22.7	DB	-31.8	94.8	5.1	3.6	-35.4	97.3	-38.4	99.4	-41.3	101.4	-45.1	104.0	
			WB	-31.9	71.6	5.1	2.3	-35.6	73.3	-38.6	74.6	-41.5	75.9	-45.2	77.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	37.1	2.8	4.8	19.5	41.2	56.0	64.9	66.5	64.3	53.9	39.5	22.0	7.9	(6)
(7)		DBStd	25.76	13.08	12.38	11.33	11.16	8.94	7.67	6.40	7.75	8.80	9.08	13.34	13.24	(7)
(8)		HDD50	6568	1464	1266	944	301	43	3	0	2	56	342	841	1306	(8)
(9)		HDD65	10497	1929	1686	1409	713	298	94	59	109	348	790	1291	1771	(9)
(10)		CDD50	1865	0	0	0	39	230	449	512	446	172	17	0	0	(10)
(11)		CDD65	320	0	0	0	0	20	90	106	89	15	0	0	0	(11)
(12)		CDH74	4066	0	0	0	15	383	1197	1206	1036	227	2	0	0	(12)
(13)		CDH80	1361	0	0	0	1	91	429	392	374	74	0	0	0	(13)
(14)	Wind	WSAvg	8.4	8.5	9.4	10.1	9.0	9.1	8.2	7.1	6.8	7.4	8.2	8.2	9.4	(14)
(15)	Precipitation	PrecAvg	14.3	0.7	0.6	0.7	1.0	1.6	1.5	2.4	1.6	0.9	1.2	1.2	0.9	(15)
(16)		PrecMax	19.5	1.5	1.2	1.5	2.6	3.9	4.2	4.8	4.5	1.8	2.2	2.1	2.1	(16)
(17)		PrecMin	8.7	0.0	0.0	0.1	0.0	0.2	0.2	0.6	0.1	0.1	0.2	0.3	0.2	(17)
(18)		PrecStd	3.0	0.4	0.3	0.4	0.7	1.0	1.0	1.3	1.3	0.4	0.5	0.5	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.8	33.1	46.0	75.7	87.4	92.2	93.4	92.8	88.1	71.0	48.9	34.1	(19)
(20)		MCWB	30.9	31.5	39.3	57.1	62.5	66.7	66.6	67.0	62.8	53.5	43.6	32.1	(20)	
(21)		2%	DB	29.0	30.2	39.2	70.2	82.2	88.6	88.4	88.4	80.4	64.4	43.9	31.8	(21)
(22)		MCWB	27.5	28.5	36.0	54.0	61.3	65.8	66.3	65.4	60.0	50.7	40.3	30.5	(22)	
(23)		5%	DB	26.0	27.2	36.4	64.9	78.3	85.2	84.2	84.3	74.4	58.8	40.0	28.6	(23)
(24)		MCWB	24.6	25.8	33.9	51.6	59.3	64.8	65.5	64.4	57.6	47.6	37.2	27.1	(24)	
(25)		10%	DB	22.6	23.5	33.9	59.5	74.2	81.0	80.8	80.1	69.2	53.7	36.5	25.5	(25)
(26)		MCWB	21.4	22.2	32.0	48.8	57.4	62.8	64.4	63.2	55.1	45.4	34.1	24.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.6	31.8	40.5	58.7	65.7	70.4	70.9	70.1	64.3	54.9	44.8	32.7	(27)
(28)		MCDWB	32.3	33.0	44.3	73.8	82.1	85.8	83.6	86.6	82.4	67.4	47.4	33.6	(28)	
(29)		2%	WB	27.7	28.5	36.3	54.9	62.9	67.8	68.9	67.5	61.0	51.7	40.6	30.4	(29)
(30)		MCDWB	28.9	29.9	38.8	67.1	78.9	84.1	81.6	83.3	77.3	62.5	43.4	31.8	(30)	
(31)		5%	WB	24.7	25.8	34.1	52.3	60.5	65.9	67.3	65.8	58.3	48.6	37.4	27.3	(31)
(32)		MCDWB	25.9	27.1	36.2	63.8	75.0	81.5	79.9	79.9	72.1	57.3	39.7	28.6	(32)	
(33)		10%	WB	21.4	22.1	32.0	49.3	58.3	64.0	65.8	64.0	56.0	45.9	34.3	24.1	(33)
(34)		MCDWB	22.6	23.5	33.9	58.8	71.7	78.2	78.0	76.9	67.8	53.3	36.2	25.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	14.1	15.0	14.8	19.1	23.2	22.9	20.7	21.8	21.3	16.7	12.4	13.9	(35)	
(36)		5% DB	MCDBR	13.9	13.3	12.3	28.9	29.8	28.8	27.0	29.1	29.3	27.2	13.9	13.3	(36)
(37)		MCWBR	13.7	13.0	10.4	15.9	14.3	12.2	11.1	11.8	14.1	16.4	11.4	13.0	(37)	
(38)		5% WB	MCDBR	13.9	13.0	11.5	25.8	26.7	25.3	21.7	25.0	26.6	23.9	13.1	13.7	(38)
(39)		MCWBR	13.8	12.7	10.0	15.3	14.2	11.7	10.5	11.7	13.6	15.3	11.3	13.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.322	0.376	0.412	0.409	0.407	0.423	0.424	0.370	0.351	0.324	0.290	(40)	
(41)		taud	2.060	2.012	1.955	2.111	2.212	2.294	2.241	2.216	2.315	2.261	2.133	2.045	(41)	
(42)		Ebn at Noon	208	238	249	258	267	268	261	253	254	230	193	181	(42)	
(43)		Edh at Noon	24	35	45	44	42	39	41	40	32	27	23	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	390	746	1209	1491	1840	1954	1819	1545	1123	600	285	258	(44)	
(45)		RadStd	28	40	124	158	149	154	138	140	82	83	36	34	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page